

Market Microstructure and Electronic Trading Finance 480

Dale W.R. Rosenthal¹

University of Illinois at Chicago,
College of Business Administration

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¹daler@uic.edu

Trades and Traders

Recall last lecture we discussed orders and quotes:

- Different types of orders;
- Different price sensitivities for orders;
- How quotes result from orders;
- Limit order books and CLOBs; and,
- Problems with CLOBs.

We also examined the famous Christie and Schultz “odd-eighth quote avoidance” paper — and did a little statistical analysis.

Readings on Today's Topic: Trades and Traders

- Chapter 2 of Johnson.
- Chapters 1–2 of Hasbrouck (previously).
- Blackboard: Reading on PKO.

Introduction

- Today we'll discuss when orders agree and a *trade* happens.
- Some trade details are order details about post-trade handling.
- Researchers use trade prices for investment, trading inferences.
- Thus trade details/data issues are crucial for doing research.
- We'll also talk about who these traders are, why they trade.
- Also discuss how they trade, how that affects the market.

Trades

- If orders express want, trades are the satisfaction of that want.
- Trades happen if buyers' and sellers' orders agree on a price.
- At that point, a few questions have to be answered:
 - What price to agree on?
 - What quantity (e.g. number of shares) to agree on?
 - Which orders get filled first?
- Affected buyers and sellers are told their orders were *filled*.
- Buyers and sellers *confirm* the trade details.

Matching Orders

When orders match on price, one or more trades happen.

A few details must be resolved:

- Price overlaps: Buyer willing to pay more than seller wants.
 - Usually trade at price of liquidity provider.
 - Exception: Discretionary orders.
- Amount inequalities: Trade maximum amount possible.
- Prioritization: usually price-time priority.
 - “Early bird gets the worm.”
 - Creates an incentive to provide liquidity.

After the Trade is Made

A few details govern what happens after a trade occurs:

- Settlement instructions;
- Delivery instructions;
- Storage needs.

Settlement

- Money and securities rarely transfer immediately after a trade.
- Money moves from buyer to seller at *settlement*.
- USEq, UST, JPEq: $T+3$; UKEq: $T+2^2$, Futures: $T+0^3$
- Most markets moving toward same-day ($T+0$) settlement.
 - Problem: Confirms can involve a lot of manual intervention.
- Can specify non-standard settlement, but:
 - These are not “regular way” orders/trades.
 - Thus they are part of a different, less-liquid market.
- Settlement usually occurs at the time of delivery.

²CREST fines participants if $<98\%$ of trades settle by $T+1$.

³Why?

- Securities move from seller to buyer at *delivery*.
- Can specify non-standard delivery date.
 - Again, this removes you from the liquid regular way market.
- Sometimes can specify different settlement, delivery dates.
 - The market for these orders is highly illiquid;
 - Same as an unsecured, no-doc, no-questions-asked loan.

- Delivery in many markets has been simplified:
 - Cash settlement of futures; and,
 - Electronic movement of securities.
- However, delivery can prove difficult:
 - Some futures require physical delivery; and,
 - Squeezes can occur (esp. naked shorts, bond futures).

Delivery: Wish You Were Here

- If physical delivery is required:
 - Coal in WY is useless for delivery to Gary, IN tomorrow.
- Even if a commodity is cash settled:
 - To deliver coal to Gary, you'll need to buy it from nearby.
 - There could be a local squeeze (little supply locally).
 - This may affect the cash market and thus cash settlement.
- These markets can see major price swings (esp. in *local basis*⁴)

⁴Difference between local price and overall cash price.

Why Failing to Deliver is Bad

- If delivery cannot be made, there is a *fail*.
- Failing may have serious repercussions:
 - Steel foundries shut down;
 - Manufacturing plants idled;
 - Power plants taken offline.
 - Workers still get paid.
- Litigation to recover economic damages becomes likely.
- Even for electronic delivery, buying someone in is a pain.

Why Failing to Deliver is Good

- Failing to deliver securities is usually less serious.
- Sometimes, fails of securities are desirable (for the buyer):
 - Same economic exposure.
 - However, need not settle until securities delivered.
 - Continue to earn interest on notional.
- *Liar's Poker*: Salomon preferenced clients likely to fail.
- Earn the float; but, you take on increased counterparty risk.

- With physical delivery, storage is also an issue.
- Delivery may merely transfer ownership of warehoused goods:
 - Gold bars in NY Fed vault;
 - Metal bars/plastic pellets in an LME warehouse.
- Storage lets speculators provide liquidity during squeezes.
- Storability duration varies wildly:
 - Metals: centuries–infinite;
 - Crude: years;
 - Grain: months;
 - Livestock: days–weeks;
 - Electricity: seconds–hours.
- Storage entails a cost, perhaps steep.

We often use trade data for inference.

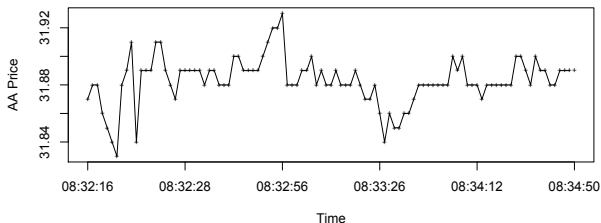
Therefore, we should be aware of some issues:

- Bid-ask bounce;
- Trading inside and outside the spread;
- Non-regular-way trades;
- Double counting of trades;
- Negotiated trades.

Some terminology. Revealing trade to public also called:

- “Printing” the trade; or,
- “Sending it to the tape.”

Bid-Ask Bounce



Alcoa Aluminum on 20070124 (Chicago Time)

- Perhaps the best-known microstructure effect.
- Due to interspersal of trades happening at static bid and ask.
- Classic newbie mistake: Assuming this return can be traded.
- Related: Not subtracting spread(s) from expected returns.

Trading Inside and Outside the Spread

- Trades can occur inside and outside the spread.
- Trades inside the spread:
 - Orders which cross at a market center (rare);
 - Internalized orders which cross;
 - Odd-lot orders inside the spread⁵;
 - Discretionary (“hidden price”) orders.
- Trades outside the spread:
 - Trades at non-NBBO market’s bid/ask;
 - Sweeping/clawing into the book;
 - Markups/markdowns;
 - Negotiated trades.
- Finally: Do you really know the inside market at trade time?

⁵I mentioned this last lecture.

Non-Regular-Way Trades

- Non-standard trades occur outside the regular-way market:
 - Non-standard delivery and/or settlement.
- These trades are usually marked as non-standard:
 - Delivery/settlement may be marked (e.g. “2” for T+2);
 - Trade may be marked as “cash⁶” (T+0)
- Must remove these for inferences about regular-way market.

⁶Not to be confused with the cash market.

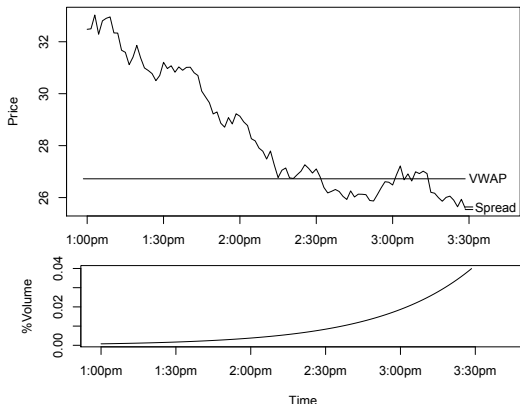
Double Counting

- Some markets require both buyer and seller to print trades.
- Thus trades appear multiple times on the tape.
 - *Uwaga!* Transaction databases may be copies of the tape.
- Especially troublesome when brokers are involved:
 - Buyer, seller, and broker (or more) may all print same trade.
- Awareness of this is crucial for inferences about volume.

Negotiated Trades

- Some prints are negotiated trades.
 - Typically large trades (otherwise, use continuous market).
 - Price negotiated to account for liquidity, adverse selection risks.
- Negotiated trades are *not* accessible from continuous market.
- Negotiated trades include:
 - block trades ($\text{int}(G127/100) \in \{1,3\}$ or COND="J" in TAQ);
 - some auction trades (tricky; varies);
 - PRP trades (COND="P" in TAQ);
 - VWAP/average price trades (COND="W" in TAQ).
- Negotiated trades can occur far outside the spread.
- Must remove these for most inferences.

How Negotiated Trades Print Outside the Spread



- VWAP sell traded from 13:00–15:30 biases price downward;
- VWAP trade printed after 15:30 at VWAP — above spread;
- Trade appears to be a buy, but is a sell.
- Print includes 13:00–15:30 trade volume (double counting)
- Has polluted many studies.

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Traders

Traders in financial markets largely fall into a few groups:

- Intermediaries (broker-dealers);
- Hedgers;
- Producers;
- Investors;
- Speculators and arbitrageurs; and,
- Governmental and quasi-governmental agencies.

- Market intermediaries help match up buyers and sellers:
 - *Brokers* pair up buyers & sellers; take no position.
 - *Dealers* trade w/buyer (seller); try to get back to 0 position.
 - Thus dealers match up buyers and sellers across time.
- *Market makers*: dealers; always quoting to buy, sell (two-sided).
 - Make money from people paying the spread, but take non-zero position.
 - Lose money when market moves against their position.
- Thus market makers can be very risk averse:
 - May only trade larger size with clients they trust; and,
 - Typically close out positions by end of day.

- *Hedgers* trade to hedge risk they don't want:
 - Utilities may hedge price of coal or natural gas;
 - Starbucks hedges price of coffee;
 - Banks hedge against interest rate changes;
 - Option holders hedge against underlying price movement.
- Hedgers may be less price-focused:
 - Hedging may be more crucial than “quibbling over cents.”
- Typically hold hedge positions for long times.
- May occasionally/regularly alter hedge position.

- *Producers* include many sellers of goods:
 - Farmers (grain);
 - Mine operators (ore/refined metal);
 - Well operators (oil/gas);
 - Listed firms (treasury stock).
- Producers differ from hedgers in that they hold *real options*:
 - Can decide to produce less/more — albeit with a time lag.
- Thus producers can exert real control on supply:
 - Gives producers valuable private information; but,
 - Listed firms' shareholders may fight dilution.

- Investors include many different groups:
 - Retail: you and me in our PAs.
 - Institutional (portfolio managers):
 - Insurers and reinsurers;
 - Pension funds and endowments; and,
 - Mutual funds, hedge funds, and managed futures.
 - Private equity: very concentrated and hands-on.
- Institutions may have mandate: only hold “vetted” securities.
 - Investment-grade bonds;
 - Indexed stocks.

Institutional Investors: Mandate Effects

Mandates on institutional investment can create liquidity events.

- Entering/leaving a vetted class can be a major event:
 - Changes universe of possible holdings (investment-grade?).
 - Some investors hold “the index” (no ifs, ands, or buts).
 - Buying/selling may be concentrated at these change times.
- Mutual funds may invest⁷ only at month-end (smaller event).
- Funds which report holdings may temporarily alter them:
 - Done around reporting time.
 - Controls what positions are reported (“window dressing”).

⁷Or sell if they have lost investors.

Speculators and Arbitrageurs

- *Speculators* take risk to make money;
- *Arbitrageurs* take no risk to make money.
 - Arbitrage is rare;
 - Some who think they are arbitrageurs are really speculators.
- Trade for profit only.
- Usually have no material interest in security *per se*.
- Includes hedge funds and CTAs⁸
- Position holding time ranges from seconds to weeks to years.
- Usually much more liquidity-savvy than other traders.
- Recall: dealers take positions to earn spread; fear market moves.
 - Thus many dealers model market; think about why prices move.
 - This often leads them to learn about market and speculate.

⁸Commodity trading advisers.

Governmental and Quasi-governmental Agencies

- Governmental/quasi-governmental agencies may trade in markets.
- Usually results in major liquidity events.
- Often trade to achieve policy objectives.
- Thus may *want* to move prices. (VERY unusual!)
- Policy objectives may reduce market efficiency.
- Examples of agencies in market:
 - US: Fed OMOs, QE, Op. Twist; FNMA/FHLMC;
 - Japan: BoJ; Price Keeping Operations (1990s+);
 - Singapore, China, etc.: Government Investment Companies;
 - China: “National Team” to support market in 2015 summer.

We've talked about market macrostructure; orders and quotes; trades and traders.

We are done with the basics! On to Market Structures.

- Market Structures: Continuous Markets; Decentralization
- Metrics/Models: Prices; Sizes; Limits; Time; Liquidity
- Electronic Trading: Markets; Tools; Tactics; Strategies
- Microstructure and Policy Issues

Readings For Next Lecture

Readings, in order of importance:

- Chapter 3 of Johnson.
- Blackboard: Readings